

Validation of MODIS Level-2 Sea Surface Temperature Using Argo Float Observations

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Abstract

This study validates Moderate Resolution Imaging Spectroradiometer (MODIS) Level-2 sea surface temperature (SST) retrievals against in-situ Argo float measurements in the western North Atlantic. Using strict spatial (10 km) and temporal (± 7 days) collocation around July 15, 2024, 12 high-quality matchups were analyzed. Results show strong agreement ($r = 0.839$, $R^2 = 0.704$) and a minimal, non-significant mean bias (-0.033°C , $p = 0.760$), consistent with the physical cool-skin effect rather than systematic sensor error. Absolute errors ($\text{RMSE} = 0.639^\circ\text{C}$; $\text{MAE} = 0.496^\circ\text{C}$) are primarily attributed to sub-grid variability within the Gulf Stream frontal zone. Overall, MODIS demonstrates strong reliability in capturing regional SST variability, supporting its use in climate studies of ocean heat transport and North Atlantic storm dynamics.

1. Introduction

Sea surface temperature (SST) is one of the most important variables for understanding ocean dynamics and its interaction with the atmosphere. SST directly governs the exchange of heat, moisture, and momentum across the air-sea interface, or the boundary layer where the atmosphere meets the ocean surface. SST data is widely used in weather forecasting and climate research (Saleh & Al-Anzi, 2021), especially as temperature effects of climate change become more extreme. The western North Atlantic is a particularly important region to study because it is dominated by the Gulf Stream, a powerful western boundary current characterized by intense thermal fronts and massive heat transport. These surface temperature gradients force the overlying boundary layer and can trigger atmospheric convection in the summertime. Massive updrafts of moist air disrupt upper-level circulation, generating forced planetary Rossby waves that interact with the permanent pressure systems of the North Atlantic and alter the pressure gradient between the two. Because this region dictates everything from regional marine ecosystems to the trajectory of winter storm tracks across the hemisphere, obtaining highly accurate, localized SST measurements here is essential.

Today, satellites are the primary tool for daily global SST monitoring because they provide much better spatial and temporal coverage than traditional ship or buoy measurements in the physical ocean (Hu et al., 2024). One of the most relied-upon instruments for this is the Moderate Resolution Imaging Spectroradiometer (MODIS-AQUA), which has been a trusted tool for ocean observation since the early 2000s (Jonasson et al., 2023). MODIS retrieves specific daytime and nighttime measurements. While nighttime retrievals can use a wider range of infrared bands, daytime retrievals must avoid shorter-wavelength channels that are easily contaminated by reflected sunlight. Therefore, daytime MODIS algorithms, such as the Non-Linear SST (NLSST) formulation, rely exclusively on thermal infrared energy from the 11 and 12 μm wavelength channels (Szczo drak & Minnett, 2022). These specific wavelengths sit within an "atmospheric window," meaning the atmosphere is relatively transparent to them. However, atmospheric water vapor still absorbs a fraction of this outgoing thermal energy. Because water vapor absorbs infrared radiation slightly differently at 11 μm than at 12 μm , the NLSST algorithm calculates to partially correct for atmospheric moisture that might otherwise absorb the ocean's heat signal before it reaches the satellite.

Despite their immense spatial coverage and advanced retrieval algorithms, satellite measurements still contain inherent uncertainties. Thermal infrared sensors like MODIS can only measure the ocean's "skin" temperature, which is the microscopic surface layer less than a millimeter thick. This skin temperature can behave differently than the water just a few meters below, and the satellite's signal can easily be distorted by clouds, aerosols, or atmospheric water vapor (Szciodrak & Minnett, 2022). Additionally, satellite data is subject to a spatial footprint effect. A single pixel on a MODIS map represents an average temperature over a large 1 km spatial area. Averaging over this footprint can obscure the sharp, localized thermal fronts characteristic of the North Atlantic (Boussidi et al., 2019). These limitations demonstrate that satellite data cannot be assumed to be perfectly consistent and flawless. MODIS SST numbers must be rigorously validated using independent, in-situ instruments to quantify any inherent measurement biases (Saleh & Al-Anzi, 2021).

To provide this necessary validation, oceanographers heavily rely on the established global Argo network, containing nearly 4,000 active floats deployed across the world's oceans. Argo floats are autonomous, battery-powered instruments that drift through the ocean, periodically diving down and then rising to the surface to measure high-resolution temperature profiles, among other variables. Because of its spatial distribution and rigorous data quality control, the Argo network is widely considered the gold standard for in-situ oceanographic data. Unlike satellites that measure the skin of a wide area, Argo floats provide highly accurate, localized measurements of the ocean's "bulk" temperature a few meters below the surface. Recent comprehensive reanalyses of satellite data have demonstrated that comparing these two different types of measurements is one of the most reliable ways to ground-truth satellite algorithms (Jonasson et al., 2023)..

In this project, I validate satellite SST retrievals by comparing MODIS SST data against near-surface temperatures from Argo floats in the western North Atlantic. By extracting data from a specific regional box and matching the satellite pixels spatially and temporally with Argo float locations, this study aims to answer the following core science questions:

- How well does MODIS SST agree with in-situ Argo temperature measurements in the western North Atlantic?
- What are the specific bias and error statistics between these two observational platforms?
- Does MODIS accurately capture the magnitude of spatial SST variability observed by the Argo floats?

2. Data and Methodology

2.1 Data Sources

To evaluate satellite performance in the western North Atlantic, I employed MATLAB to analyze two primary datasets centered around a target date of July 15, 2024. The mid-summer period was selected to maximize clear-sky observations and favorable atmospheric conditions for infrared SST retrievals, while also capturing peak upper-ocean stratification and strong temperature gradients.

MODIS Level-2 Sea Surface Temperature: I analyzed MODIS Level-2 (L2) SST data rather than Level-3 (L3) data to allow for the most accurate spatial matchup with localized Argo float coordinates. L3 products are gridded and temporally averaged, an interpolation process that smooths out the fine-scale spatial gradients of the ocean surface. Ungridded L2 swath data preserves the original 1 km spatial resolution of the satellite pixels in a region dominated by sharp thermal fronts. I obtained the data via the NASA OceanColor Web (Direct Data Access) portal. I selected the files corresponding to July 15, 2024 pixels and a regional bounding box in the western North Atlantic (30°N to 45°N, 75°W to 45°W). The files were downloaded in NetCDF (.nc) format, which contains the raw pixel-level SST values calculated using the Non-Linear SST (NLSST) algorithm. To ensure data reliability, only pixels with a designated quality control flag of 1 or lower ($\text{qual_sst} \leq 1$) and realistic temperature values were retained for analysis. A flag of 0 or 1 represents high quality data values that have passed critical consistency tests. Flags of 2 or higher are often contaminated by sub-pixel cloud cover or aerosols.



Figure 1: Western North Atlantic Region Selected (source: NASA Ocean Color Map)

Argo Float Profiles: I acquired in-situ temperature validation data from the global Argo float network by accessing profiles from the IFREMER Argo Global Data Assembly Centre (GDAC). Profiles were filtered to only extract temperature measurements within the top 10 decibars, or 10 meters, of the water column. In the upper ocean, the density of seawater is relatively constant near the surface, so I treated pressure in dbar as numerically equivalent to depth in meters. These shallow temperature values were taken from the PRES_ADJUSTED and TEMP_ADJUSTED Argo variables. These have undergone quality control and passed Real-Time Quality Control (RTQC) tests to correct for potential float drift or pressure sensor offsets. Even within the adjusted variables, only measurements flagged as "good" (QC = 1) or "probably good" (QC = 2) were retained to establish the best data for validation. A temporal window of ± 7 days from the target date of July 15, 2024, was applied to capture sufficient in-situ profiles in the region. This temporal window balances the need for sufficient matchups with the rapidly evolving nature of ocean surface conditions.

2.2 Coupling Procedure

To directly compare the satellite retrievals with the in-situ measurements, I executed a spatial "point-to-pixel" collocation procedure. Because an Argo float provides a single-point measurement and the MODIS satellite provides a grid of pixels, the two datasets had to be mathematically aligned in space.

For every valid Argo profile, I used a search algorithm to calculate the great-circle distance (the shortest distance over the Earth's curved surface) between the float's coordinates and every available MODIS pixel in the region. A strict spatial threshold of 10 km was established to define a successful "matchup." If the center of the nearest MODIS pixel fell within this 10 km radius of the Argo float, the two values were paired for comparison. This 10 km limit was chosen to minimize the risk of comparing water masses with different thermal properties, especially near the sharp gradients of the Gulf Stream. This filtering process successfully yielded 12 high-quality MODIS-Argo matchups for statistical evaluation.

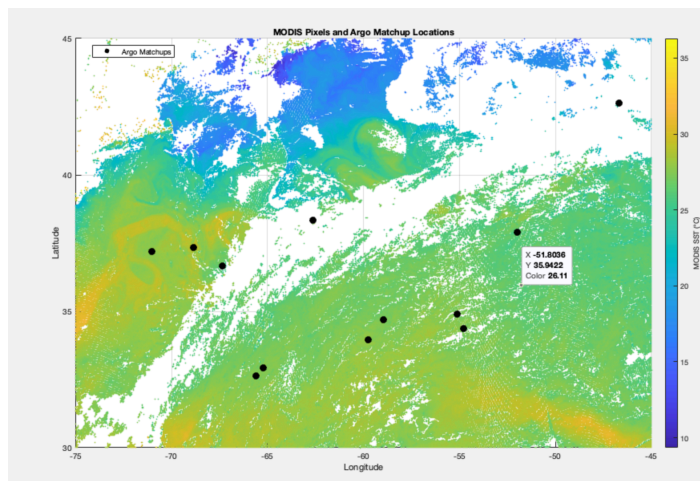


Figure 2: 12 MODIS-Argo SST Matchup Locations and SST. Sea surface temperatures in the study region typically ranged from approximately 20°C to 30°C, consistent with expected summer conditions in the western North Atlantic.

2.3 Statistical Analysis

I evaluated the accuracy and precision of the MODIS SST retrievals against the Argo baseline using core statistical metrics. The mean bias is helpful in determining systematic overestimation or underestimation by the satellite. The Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) quantify the absolute variance and average magnitude of the MODIS retrieval errors. To assess the linear agreement between the two datasets, I calculated a Pearson correlation coefficient (r) and a coefficient of determination (R^2). Finally, I used a paired t-test on the matchups to determine if the calculated mean bias was statistically significantly different from zero at the 95% confidence interval.

3. Results

3.1 Linear Regression and Figures

The linear regression analysis of the 12 matchups yielded the following metrics:

- Pearson Correlation Coefficient: $r = 0.839$
- Coefficient of Determination: $R^2 = 0.704$
- Regression Equation: $\text{MODIS} = 1.077(\text{Argo}) - 2.110$
- Correlation p-value = 1.6×10^{-10}

The statistical agreement between the MODIS Level-2 SST and the in-situ Argo measurements is high. The Pearson Correlation Coefficient of 0.839 indicates a strong positive linear relationship. This is further supported by the Coefficient of Determination of 0.704, meaning that approximately 70.4% of the variability observed in the MODIS SST data is directly accounted for by the variability in the Argo in-situ measurements. As illustrated in Figure 3 below, the matchup points track closely along the 1:1 line of perfect linear agreement. The resulting regression equation suggests similar: $\text{MODIS} = 1.077(\text{Argo}) - 2.110$, though there is a slight deviation from the 1.0 slope. The linear fit proposes that for every 1°C ocean warms up (measured by Argo), the MODIS satellite registers a 1.077°C increase, and if the Argo float measured an ocean temperature of exactly 0°C, the MODIS satellite would theoretically read -2.110°C. Unsurprisingly, I found the correlation to be statistically significant with the p-value = 1.6×10^{-10} (<0.05).

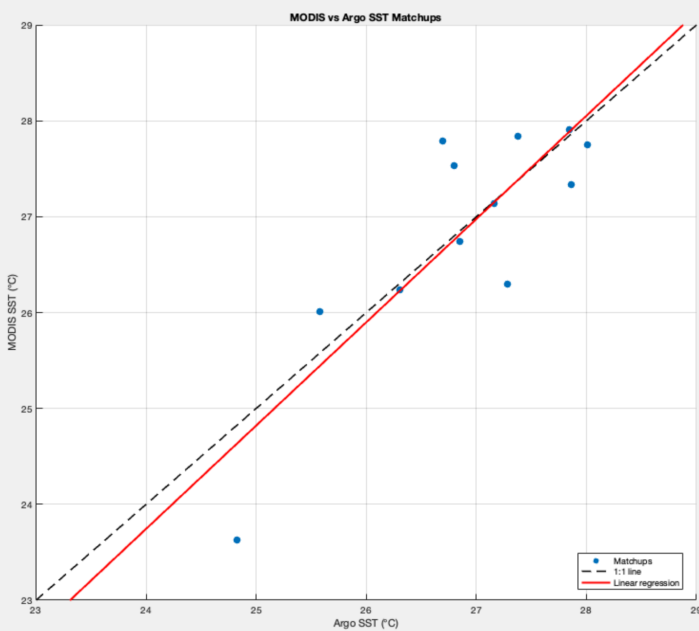


Figure 3: Correlation Between MODIS and Argo SST

Scatter plot of MODIS Level-2 SST versus in-situ Argo bulk temperature. The dashed black line represents the 1:1 line of perfect agreement, while the solid red line indicates the linear regression fit ($R^2 = 0.704$).

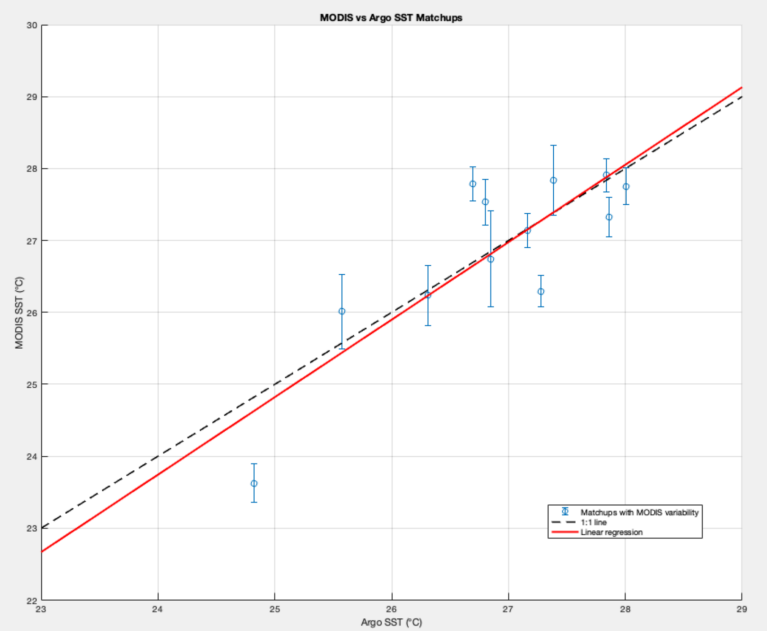


Figure 4: MODIS vs. Argo Matchups with Local Variability

Figure 3 with error bars. The error bars represent the standard deviation of all valid MODIS pixels within the search radius of each Argo float.

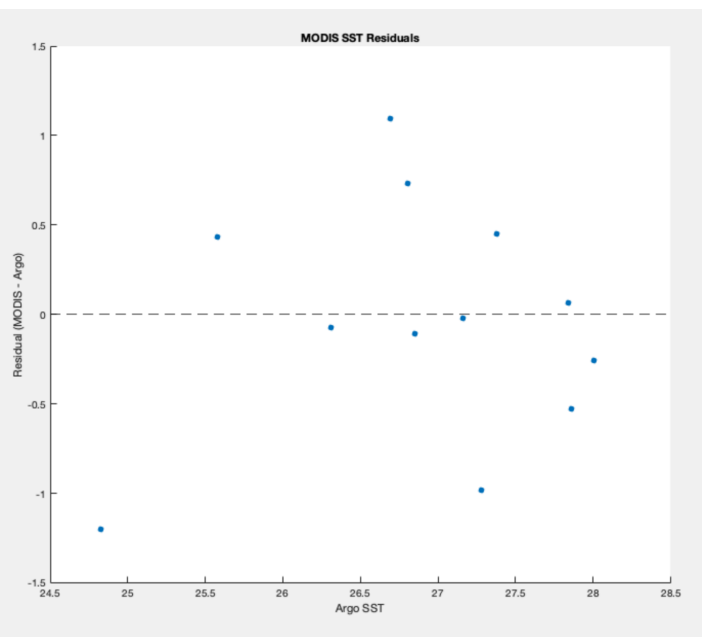


Figure 5: Residual Analysis of MODIS SST Retrievals
Scatter plot of the residuals (MODIS_SST - Argo_SST) against the in-situ Argo temperature measurements. The dashed horizontal line at 0 degrees Celsius represents the line of zero bias.

3.2 Error and Bias Statistics

The accuracy and precision error metrics of the matchups are:

- Mean Bias (MODIS - Argo) = -0.033°C
- Root Mean Square Error (RMSE) = 0.639°C
- Mean Absolute Error (MAE) = 0.496°C
- Paired t-test p-value = 0.760
- 95% Confidence Interval (CI): $[-0.252, 0.186]^{\circ}\text{C}$
- t-statistic $t(11) = -0.307$

The Mean Bias of -0.033°C reveals that the MODIS sensor recorded slightly cooler temperatures, on average, than the Argo floats. However, as shown in the residual plot (Figure 5), the individual differences are distributed evenly above and below the zero line. This distribution verifies that the errors are primarily random rather than systematic. The RMSE (0.639°C) and MAE (0.496°C) quantify the magnitude of these individual variations. Specifically, the MAE demonstrates that the MODIS SST typically deviates from the in-situ measurement by approximately $\pm 0.5^{\circ}\text{C}$. The higher value of the RMSE relative to the MAE reflects the mathematical sensitivity of the RMSE to larger errors, confirming the presence of a few high-magnitude outliers within the matchup dataset.

The spatial variability contributing to these differences is illustrated in Figure 4. The vertical error bars represent the standard deviation of the MODIS pixels within the 10 km search radius of each float. The varying sizes of these error bars reflect local sea surface variability, showing that some matchups occurred in uniform waters while others contained higher local temperature variations.

Finally, a paired t-test ($n = 12$) was performed to determine the statistical significance of the bias. The resulting t-statistic was $t(11) = -0.307$ with a p-value of 0.760. Because the p-value is greater than 0.05, the mean bias is not statistically significant. Additionally, the 95% Confidence Interval of $[-0.252, 0.186]^{\circ}\text{C}$ includes zero, validating that a true mean difference of zero is statistically possible. Together, these metrics indicate there is no systematic over- or underestimation of SST by the MODIS sensor in this study area.

3.3 Data Table

Matchup ID	Distance (km)	Time Difference	Residual Error ($^{\circ}\text{C}$)
1	9.27	5.24	-1.201
2	0.71	-1.49	0.453
3	1.84	-5.72	-0.106
4	6.15	5.25	-0.985
5	1.51	-2.37	-0.073
6	0.4	-3.93	0.733
7	0.45	6.09	-0.257
8	0.6	-3.48	-0.023
9	0.29	6.52	0.064
10	0.76	-6.84	1.094
11	0.78	3.2	-0.529
12	4.05	-4.35	0.432

Table 1: Spatial and Temporal Differences for all 12 Matchups

(Note: A negative Time Difference in days means the Argo float measured the water before the satellite target date; a positive difference means it measured it after. A negative residual means the MODIS satellite reads colder than the Argo float.)

5. Discussion

The statistical evaluation of MODIS SST against in-situ Argo profiles in the North Atlantic yielded a Mean Bias of -0.033°C , an RMSE of 0.639°C , and an MAE of 0.496°C . Rather than indicating sensor malfunction, these metrics reflect a combination of fundamental oceanography patterns, spatial variability, and limitations inherent to satellite remote sensing.

5.1. Sensor Physics: The Skin vs. Bulk Temperature Effect

The overall negative Mean Bias is physically consistent with the vertical thermal structure of the upper ocean. MODIS is a satellite infrared radiometer, meaning it measures the electromagnetic radiation emitted from the ocean's "skin" layer. The "skin" SST is a proxy for the temperature at approximately a 1 mm depth, while Argo floats measure the "bulk" temperature at depths typically starting around 5 meters.

Because the ocean surface is subjected to continuous heat loss to the atmosphere through evaporation, sensible heat flux, and net longwave radiation, the skin layer is persistently cooler than the underlying bulk water (Szciodrak & Minnett, 2022). Oceanographers establish that the skin layer is typically 0.1°C to 0.5°C cooler than the bulk under moderate wind conditions. This physical theory is supported by Matchup 4 (33.56°N , 58.76°W), which recorded a residual of -0.204°C . Falling within the expected range, this matchup and the observed mean bias is a likely observation of the cool skin effect within the study area.

5.2. Sub-Grid Variability in the Gulf Stream Region

While the Mean Bias is minimal due to the cancellation of positive and negative residuals, the RMSE 0.639°C and MAE (0.496°C) reveal substantial individual discrepancies. These magnitudes are expected given the geographic context of the study, as the western North Atlantic is heavily influenced by the Gulf Stream.

In such frontal zones, temperature gradients can change by several degrees over just a few kilometers at the surface. A 10 km spatial window in this region means that a satellite pixel and an Argo float may be measuring entirely different water masses. For instance, one situated in a warm core ring and the other in the cooler surrounding shelf water. The variable vertical error bars in Figure 4 confirm this sub-grid variability.

5.3. Influence of Spatial and Temporal Collocation

The limitations of matching dynamic ocean features are further quantified in the individual matchup statistics (Table 1). The accuracy of the validation is fundamentally constrained by "representative error," or the natural environmental variance that occurs during the gap between the satellite overpass and the float surface.

This relationship is explicitly visible in the data. Located in a stable water mass, Matchup 8 (37.07°N , 60.18°W) exhibited a near-zero residual -0.023°C , driven by a tight spatial arrangement of 0.60 km. On the other hand, Matchup 1, which contained the maximum spatial separation (9.27 km), produced the largest negative error of -1.201°C . Situated in the highly dynamic Slope Water region near the Gulf Stream's North Wall, the larger separation in such a frontal zone may have **resulted** in the two instruments sampling different water masses. The dataset also features the danger of temporal drift; Matchup 10 was physically close (0.76 km) but separated by nearly 7 days, resulting in a $+1.094^{\circ}\text{C}$ error due to the constant evolution of the local water mass. These specific outliers drag the RMSE upward and emphasize the necessity of strict space and time thresholds in advective environments.

5.4. Atmospheric Attenuation and Algorithm Limitations

The slight negative skew of the dataset may be partially attributed to seasonal atmospheric conditions. The standard Non-Linear SST (NLSST) algorithm utilizes the differential absorption between the $11\text{ }\mu\text{m}$ and $12\text{ }\mu\text{m}$ brightness temperature bands to estimate and correct for water vapor attenuation (Jonasson et al., 2023).

During July in the North Atlantic, atmospheric moisture is typically at its peak. In highly humid environments, the empirical coefficients of the NLSST algorithm can sometimes undercompensate for the severe attenuation of the surface signal by atmospheric water vapor. Consequently, the satellite sensor receives less infrared energy, which the algorithm interprets as a colder surface temperature. This atmospheric interference likely acts in tandem with the physical cool skin effect on the mean bias.

5.5. Seasonal Stratification and the Diurnal Thermocline

The mid-July timing of this study introduces specific physical considerations related to summer stratification. During this time in the North Atlantic, intense solar insolation combined with typically lower wind speeds leads to the formation of a "diurnal thermocline." When solar heating is strong, the temperature gradient in the top few meters of the ocean becomes distinctly non-linear.

While the "cool skin" effect usually makes the surface colder than the bulk, intense, summer daytime heating can actually cause the skin layer to become warmer than the water at the 5-to-10-meter depth measured by Argo floats. This phenomenon, known as diurnal warming, can create a "warm layer" that masks the underlying bulk temperature. Because the matchups in this study occurred during this peak heating window, the observed mean bias of -0.033°C is likely a residual sum of two competing forces: the persistent physical cool skin effect and the intermittent summer diurnal warming. The fact that the bias is so close to zero suggests that, in the North Atlantic during July, these two effects may partially offset one another, a nuance that global validation studies often overlook.

5.6. View Angle and Atmospheric Path Length Effects

The accuracy of L2 swath data is also intrinsically linked to the geometry of the satellite's observation. Unlike Level-3 gridded products, L2 data retains the original "Sensor Zenith Angle" (SZA), or the angle at which the MODIS instrument views the ocean surface. As the view angle increases toward the edge of a satellite swath, the "path length" (the distance the infrared signal must travel through the atmosphere) increases significantly.

A longer path length through the humid July atmosphere of the North Atlantic results in greater signal attenuation by water vapor and aerosols. This "limb darkening" effect can lead to cooler-than-actual temperature retrievals at the edges of the swath. Given that this study utilized ungridded L2 data to preserve spatial resolution, some of the higher-magnitude residuals observed in Table 1 (such as Matchup 1) may be the result of the float being positioned near the edge of a MODIS swath. This geometric factor contributes to the observed RMSE of 0.639°C and is an example of a limitation of using single-swath L2 data over temporally averaged products.

5.7. Algorithm Sensitivity and Regional Benchmarking

The MODIS SST retrieval algorithm method is inherently tied to the "training sets" used to derive its coefficients. NLSST is an empirical algorithm, meaning its ability to correct for atmospheric moisture is only as good as the global data used to calibrate it. In regions with unique atmospheric profiles, such as the North Atlantic's humid boundary layer, the global coefficients may not be perfectly tuned.

My RMSE of 0.639°C and MAE of 0.496°C fall within the typical range compared with previous MODIS vs. in-situ SST validation studies reporting RMSE values around $0.5\text{--}0.7^{\circ}\text{C}$. This suggests that while traditional retrieval models are "not fully applicable for offshore waters where conditions change a lot" (Hu et al., 2024), the MODIS NLSST algorithm performs reliably in the Western North Atlantic, possibly even compared to other global ocean basins. The high R^2

of 0.704 achieved in this study reveals that despite the "thermal noise" of the Gulf Stream, the satellite can remain a mostly accurate tool for capturing regional temperature variance.

5.8. Implications for North Atlantic Climate Modeling

Finally, the validation of MODIS SST in this specific region has significant implications for the atmospheric circulation patterns described in the introduction. As established by Cassou et al. (2004), the "North Atlantic Horseshoe" (HS) SST mode during the summer is a primary driver of the North Atlantic Oscillation (NAO) in the following winter.

The ability of MODIS to capture these summer SST anomalies with a low systematic bias ensures that the "forced planetary Rossby waves" triggered by these temperature gradients are being modeled based on accurate initial conditions. If the satellite were to significantly underestimate the Gulf Stream's warmth, climate models would fail to predict the subsequent shifts in the jet stream and winter storm tracks across the Northern Hemisphere. Therefore, the high correlation ($r = 0.839$) and low bias found in this study provide confidence in using MODIS L2 data for long-term climate forecasting and boundary layer interaction studies in the North Atlantic.

5.9. Study Limitations

A key limitation of this study is the small sample size ($n = 12$), which reduces confidence in the statistical estimates and limits broad application of the results of this study. In particular, the paired t-test assumes that the differences are approximately normally distributed, an assumption that is difficult to verify with a limited number of matchups. As a result, the statistical conclusions, including the non-significant bias, should be interpreted with caution.

6. Conclusion

Understanding the thermal dynamics of the ocean-atmosphere boundary layer is critical for modern climate research, particularly in advective regions like the western North Atlantic. The primary objective of this study was to validate MODIS Level-2 Sea Surface Temperature (SST) retrievals against high-resolution, in-situ bulk temperature profiles from the global Argo float network. By temporally and spatially coupling ungridded satellite pixels with Argo profiles centered around July 15, 2024, my project evaluated the precision and reliability of the MODIS Non-Linear SST (NLSST) algorithm in a region characterized by thermal fronts and high summer atmospheric humidity.

The statistical evaluation of the 12 resulting matchups suggests that the MODIS sensor performs with accuracy. The slight negative mean bias was insignificant, and more so acted as a physical observation of the ocean's theoretical cool-skin layer. Random, location-dependent variance was also noted as spatial heterogeneity rather than inherent algorithmic flaws. MODIS appears to capture a substantial portion of the observed temperature variability in the western North Atlantic. Error statistics further support this agreement, with consistent RMSE and MAE values within previously reported MODIS validation ranges, though RMSE is on the higher end ($>0.5^{\circ}\text{C}$).

However, my interpretations are subject to several methodological limitations. The strict quality control and filtering yielded a small sample size, and the constraints led to spatial and temporal mismatches. A 10 km search radius is wide enough to cross distinct boundaries in frontal zones (e.g., Matchup 1), and a 7-day temporal window is long enough to encompass significant advective drift by the Gulf Stream (e.g., Matchup 10). The higher RMSE is likely due to a combination of true sensor error and the natural evolution of the water masses.

Ultimately, the strong validation of MODIS Level-2 data in this region has broader implications for oceanographic and atmospheric sciences. Because the western North Atlantic Gulf Stream drives massive heat transport that subsequently influences the North Atlantic Oscillation (NAO) and hemispheric winter storm tracks, climate modelers must rely on highly accurate surface initializations. Future research should look to expand on this validation by utilizing larger, multi-season datasets and implementing much tighter temporal thresholds (e.g., 24 hours). Additionally, as suggested by recent literature, integrating Machine Learning techniques to predict and correct for the rapid advection of water masses between satellite overpasses will be essential for reducing representative error and improving global remote sensing validation efforts.

7. References

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